1 Powers of a diagonalizable operator

Exercise 1.1.

(a) Prove that if v is an eigenvector of T with eigenvalue λ and n is a positive integer, then $T^n v = \lambda^n v$.

(b) If T is diagonalizable, then there exists an invertible matrix P such that $D = P[T]P^{-1}$, where $[T]$ denotes the matrix of T with respect to the standard basis and D is the diagonal form of T (that is, it is the matrix of T with respect to a basis such that this matrix is diagonal). Prove that, for any positive integer n, $[T]^n = P^{-1}D^nP$.

Exercise 1.2. Define $T \in \mathscr{L}(F^3)$ by $T(x,y,z) = (2x+y, 5y+3z, 8z)$.

(a) Find the matrix of T with respect to the standard basis.

(b) Find a basis of F^3 with respect to which the matrix of T is diagonal and write the matrix of T in this basis.

(c) Calculate $T^{100}(0,0,1)$. (No need to simplify).

Exercise 1.3. The *Fibonacci sequence* $F_0, F_1, F_2, \ldots$ is defined by

$$F_0 = 0,\ F_1 = 1,\ \text{ and } F_n = F_{n-2} + F_{n-1} \text{ for } n \geq 2.$$

Define $T \in \mathscr{L}(\mathbb{R}^2)$ by $T(x,y) = (y, x+y)$.

(a) Show that $T^n(0,1) = (F_n, F_{n+1})$ for each nonnegative integer n.

(b) Find the eigenvalues of T.

(c) Show that T is diagonalizable and diagonalize T.

(d) Use the solution to (c) to compute $T^n(0,1)$. Conclude that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

for each nonnegative integer n.

(e) (Optional) Use part (d) to conclude that if n is a nonnegative integer, then F_n is the integer that is closest to $\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n$. *Hint: Show that* $\left|\frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n\right| < \frac{1}{2}$ *for all* $n \neq 0$ *and then* $\left|\frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n - F_n\right| < \frac{1}{2}$, *which means that* $\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n$ *rounds to the integer* F_n.